

PRAHARI

Predictive Road Analytics & Hotspot Intelligence for Enforcement

A Traffic Intelligence Platform · Beyond Detection, Towards Decision

Every camera captures violations. PRAHARI tells authorities where to act, when to act, and why it matters.

SECTION 1 — ALIGNMENT WITH EXPECTED OUTCOME

Expected Outcome (Theme 3): "A scalable AI-based traffic image analysis system capable of automatically identifying, classifying, and documenting traffic violations from photographic evidence, thereby reducing manual effort and improving the effectiveness of traffic monitoring and enforcement."

PRAHARI is purpose-built to satisfy every part of this expected outcome — and to extend meaningfully beyond it. The table below maps each requirement to a concrete PRAHARI capability.

Expected Outcome Requirement	How PRAHARI Delivers It
Scalable AI-based system	YOLOv8 inference containerised via Docker / FastAPI. Each module scales independently. Adding a new camera requires zero retraining. City-wide deployment across 47+ junctions from a single model instance.
Automatically identifying violations	7 violation classes detected without human trigger: helmet, seatbelt, triple riding, wrong-side driving, stop-line, red-light, and illegal parking. Works on static images and live video frames.
Classifying violations	Each detection gets a violation class + confidence score + severity tier (Critical / High / Medium / Low). PRAHARI additionally classifies the vehicle as First-time / Repeat / Habitual offender — extra classification that directly serves enforcement.
Documenting from photographic evidence	Every violation generates an evidence-grade packet: annotated image, bounding-box coordinates, OCR plate text, confidence score, GPS, timestamp, and a natural-language explanation of why it was flagged.
Reducing manual effort	Automated detection removes frame-by-frame human review. The Officer Deployment Optimizer auto-generates the next day's patrol schedule. The Junction Health Index replaces manual weekly reporting with a single 0–100 score per junction.

Improving effectiveness of monitoring	City-wide Violation Heatmap and Junction Health Index leaderboard give commissioners an at-a-glance view of every junction's safety status, updated in real time.
Improving effectiveness of enforcement	Behavioural DNA score tells officers which vehicle to prioritise. The 48-hour forecast tells them when violations will spike. The Deployment Optimizer tells them where to stand and why — turning detection into action.

Key insight: the expected outcome asks for two things — detecting violations, and improving the effectiveness of enforcement. PRAHARI is designed around the second. It tells the city not just what happened, but what to do next.

SECTION 2 — PROBLEM STATEMENT

Bengaluru operates over 8,000 surveillance cameras across its road network. Every day, thousands of traffic images are generated — yet only a small fraction are reviewed by human analysts. The result is an enforcement system that is simultaneously data-rich and intelligence-poor.

2.1 Why existing approaches fall short

- Detection without memory: standard systems treat each image in isolation, with no knowledge of whether a vehicle is a first-time or habitual offender.
- No enforcement direction: detecting a violation does not tell an officer where to be, when to be there, or which vehicle to prioritise.
- Reactive by design: enforcement is triggered after violations occur, never before. There is no mechanism to predict and pre-empt.
- Vanity metrics over impact: a model with mAP 0.87 sounds impressive but tells a traffic commissioner nothing about whether the city is safer.

2.2 The real problem

Bengaluru does not have a detection problem. It has an intelligence problem. The city has the cameras. What it lacks is the system that turns camera footage into enforcement decisions.

SECTION 3 — PROPOSED SOLUTION: PRAHARI

PRAHARI (Predictive Road Analytics and Hotspot Intelligence for Enforcement) is a five-layer Traffic Intelligence Platform. It begins with computer vision — satisfying the brief exactly — and extends into behavioural analytics, predictive forecasting, and enforcement decision support.

Core principle: Detection is the floor, not the ceiling. Every violation detected feeds a learning system that makes the next enforcement action smarter.

3.1 The five layers

Layer	Name	What it does	Outcome served
1	Data Ingestion	CCTV streams, static uploads, GPS metadata, historical challan records	Scalable system
2	Perception Engine	YOLOv8 detection, 7-class violation classifier, PaddleOCR plate extraction	Identifying + classifying
3	Intelligence Layer	Behavioural DNA engine, LSTM forecast model, Junction Health Index	Enforcement effectiveness
4	Decision Support	Officer Deployment Optimizer, Enforcement ROI Calculator	Reducing manual effort
5	Output Layer	City dashboard, officer mobile app, evidence packets, policy reports	Documenting violations

SECTION 4 — PERCEPTION ENGINE (CORE CV PIPELINE)

This layer directly fulfils the Theme 3 tasks as specified: image preprocessing, vehicle detection, violation detection, violation classification, license plate recognition, evidence generation, analytics, and performance evaluation.

4.1 Image preprocessing

- CLAHE histogram equalisation for night-time and low-light frames.
- Wiener-filter deblurring for motion blur from fast-moving vehicles.
- Albumentations synthetic rain and fog overlays to harden the model against adverse weather.
- Output: a standardised 640×640 tensor ready for inference.

4.2 Vehicle and road-user detection

- YOLOv8 fine-tuned for Indian road conditions.
- Detects and localises two-wheelers, three-wheelers, cars, heavy vehicles, riders, drivers, and pedestrians.
- Minimum confidence threshold of 0.50 for violation flagging to limit false positives.
- ByteTrack multi-object tracking for video frames recovers occluded detections.

4.3 Violation detection and classification — 7 classes

Violation	Detection method	Severity weight	Risk tier
Helmet non-compliance	Head keypoint + binary classifier on rider region	20	Medium
Seatbelt non-compliance	Pose estimation — shoulder/lap keypoint delta	25	Medium
Triple riding	Person count within	40	High

	two-wheeler box ≥ 3		
Illegal parking	Static vehicle in no-parking zone overlay	50	High
Wrong-side driving	Optical-flow direction vs. lane vector	80	Critical
Stop-line violation	Vehicle centroid crosses reference line at red phase	85	Critical
Red-light violation	Vehicle in motion during red signal state	90	Critical

4.4 License plate recognition

- PaddleOCR extracts raw plate text from the detected vehicle bounding box.
- An Indian-plate regex post-processor cleans the output to the standard KA-00-XX-0000 format.
- The plate is matched against the vehicle registry to return owner tier and history flag.

4.5 Evidence generation — verifiable packets

Every confirmed violation produces a structured evidence packet containing:

- Original frame (unmodified).
- Annotated frame: bounding boxes, class labels, confidence scores, timestamp overlay.
- Structured metadata: plate number, vehicle class, violation type, GPS, camera ID, confidence score.
- Natural-language explanation, e.g. "Rider on KA05AB7734 detected without helmet at 94% confidence, Silk Board Junction, 18:42 IST."
- Digital signature hash for a tamper-proof audit trail.

4.6 Performance evaluation targets

Metric	Formula	Target
Precision	$TP / (TP + FP)$	≥ 0.87 per class
Recall	$TP / (TP + FN)$	≥ 0.82 per class
F1-Score	$2 \times P \times R / (P + R)$	≥ 0.84 per class
mAP@0.5	Mean AP, IoU = 0.5	≥ 0.78 overall
mAP@0.5:0.95	COCO standard benchmark	≥ 0.55 overall
Throughput	Frames per second	≥ 15 FPS on T4 GPU
Plate OCR accuracy	Correct plates / total	$\geq 80\%$ on Indian plates

SECTION 5 — INTELLIGENCE LAYER (THE DIFFERENTIATOR)

This layer is what makes PRAHARI more than a detector. It is built on top of the core CV pipeline and amplifies it — every component maps directly to "improving the effectiveness of traffic monitoring and enforcement."

5.1 Behavioural DNA engine — vehicle-level recidivism scoring

Every vehicle that appears in a violation event is given a rolling Behavioural DNA score (0–100) computed from three factors:

Factor	Weight	Logic
Violation frequency (30-day window)	40%	More violations in recent 30 days → higher score
Severity-weighted violation sum	40%	Red-light (90) outweighs helmet (20) per event
Recency decay	20%	Violations in last 7 days weighted 3× over older ones

DNA score thresholds define the offender tier:

- 0–39: First-time offender — standard challan issued.
- 40–69: Repeat offender — challan plus a follow-up flag.
- 70–100: Habitual offender — escalated challan plus automatic alert to a senior officer.

Enforcement impact: Officers at a junction can scan a plate and instantly see the vehicle's tier. A habitual offender is intercepted and escalated immediately, rather than getting the same response as a first-time violator.

5.2 Junction Health Index (JHI) — city-wide safety scoring

The JHI is a single 0–100 composite score for every monitored junction, updated hourly. It replaces a multi-page weekly report with one number a commissioner can act on immediately.

JHI formula:

$$\text{JHI} = (\text{Violation Density} \times 0.35) + (\text{Severity-Weighted Score} \times 0.30) + (\text{Repeat Offender Ratio} \times 0.20) + (\text{Peak-Hour Frequency} \times 0.15)$$

Junction	JHI	Risk tier	Primary violation driver
Silk Board Junction	94	Critical	Wrong-side + red-light
Marathahalli Junction	81	Critical	Illegal parking + triple riding
MG Road Signal	73	High	Red-light violation
Hebbal Flyover	67	High	Helmet non-compliance
KR Puram Signal	54	Medium	Stop-line violation
Bannerghatta Road	38	Low	Seatbelt non-compliance

5.3 48-hour violation forecast — LSTM temporal model

An LSTM network trained on 90 days of per-junction violation time-series forecasts violation counts per type for each junction over the next 48 hours, adjusted for:

- Day-of-week patterns (weekday vs. weekend behaviour).
- Time-of-day signatures (each junction has its own peak-hour rhythm).
- Special-event calendar flags (festivals, matches, rallies).
- Weather feed (rain increases certain violation types).

Example output: Silk Board Junction, Monday 6–8 PM → wrong-side driving +62%, illegal parking +45%. The system raises the JHI forecast to 97 and triggers a deployment recommendation.

SECTION 6 — DECISION SUPPORT LAYER

6.1 Officer deployment optimizer

Given the number of available officers and the predicted violation heatmap, the optimizer generates the patrol schedule that maximises violations deterred per officer-hour, using a greedy allocation over the JHI forecast surface.

Input	Output
Officers available tomorrow	Officer-to-junction assignment
48-hour JHI forecast per junction	Shift timing per officer (AM / PM / night)
Habitual-offender location predictions	Priority vehicle watch-list per zone
Historical deterrence rate per junction	Expected violations deterred and challan ROI

6.2 Enforcement ROI calculator

For each proposed deployment, the system computes:

- Projected challan revenue (violation count × average challan value).
- Officer-hour cost of the deployment.
- Net enforcement ROI ratio.
- Estimated accident-cost avoidance (severity × accident probability × average accident cost).

Example: deploy 3 officers at Silk Board Junction, Monday 6–8 PM. Projected: 180 violations deterred, ₹54,000 challan recovery, ₹1,800 officer cost, ~30× net ROI, ₹2.4 lakh accident-cost avoidance.

SECTION 7 — WHY PRAHARI STANDS APART

A standard detection system answers one question — "did a violation occur?" PRAHARI answers the questions a traffic commissioner actually asks each morning: who are the worst offenders, where will violations happen, and where should officers be deployed today?

Capability	A detection-only system	PRAHARI
Violation detection	YOLOv8 — 7 classes	YOLOv8 fine-tuned — 7 classes + calibrated confidence
Vehicle memory	None — each image isolated	Behavioural DNA — rolling 30-day history per plate
Enforcement direction	None	Officer Deployment Optimizer — daily patrol schedule
Temporal intelligence	None	LSTM 48-hour forecast per junction, event-adjusted
City-level view	Bar chart of violation types	Junction Health Index — 0–100 score, live leaderboard
Evidence quality	Annotated image	Verifiable packet with NL explanation + hash signature
ROI quantification	None	Challan revenue + accident-cost avoidance per deployment
Repeat-offender handling	None	Three-tier escalation (First /

		Repeat / Habitual)
--	--	--------------------

SECTION 8 — ARCHITECTURE & TECHNOLOGY STACK

8.1 Technology stack

Component	Technology
Object detection	YOLOv8 (Ultralytics) — fine-tuned for Indian roads
Image preprocessing	OpenCV + Albumentations
Multi-object tracking	ByteTrack (video frame sequences)
OCR — license plates	PaddleOCR + Indian-plate regex post-processor
Behavioural DNA store	PostgreSQL + TimescaleDB (per-plate time-series)
Forecast model	LSTM (PyTorch)
Deployment optimizer	Greedy allocation algorithm (Python)
Backend API	FastAPI — inference + intelligence endpoints
Dashboard	React.js — live JHI leaderboard, heatmap, DNA lookup
Evidence store	MinIO object storage + PostgreSQL metadata
Training infrastructure	Google Colab Pro / Kaggle (T4 GPU)
Containerisation	Docker — each layer independently deployable

8.2 Data flow (end-to-end)

1. Image enters the Ingestion Layer (upload or RTSP stream frame).
2. Preprocessing normalises for lighting, blur, and weather.
3. YOLOv8 runs inference — bounding boxes, class labels, confidence scores.
4. Violation classifier assigns severity tier and violation class.
5. PaddleOCR extracts plate text; the regex cleaner normalises it.
6. Plate looked up in the Vehicle Behaviour DB — DNA score retrieved or initialised.
7. DNA score updated with the new event (weighted-recency formula).
8. Offender tier assigned (First / Repeat / Habitual); challan priority set.
9. Evidence packet assembled: annotated image + metadata + NL explanation + hash.
10. JHI recalculated for the junction and pushed to the live dashboard.
11. LSTM forecast refreshes its 48-hour projection.
12. Deployment Optimizer re-runs and updates the next day's patrol schedule.

SECTION 9 — SCALABILITY & REAL-WORLD DEPLOYMENT

9.1 Edge deployment

YOLOv8 exported to TensorRT runs on NVIDIA Jetson Orin edge devices co-located with cameras, cutting cloud bandwidth by around 90% and enabling sub-200ms local alerting. Only violation events — not raw video — are sent to the cloud.

9.2 City-scale horizontal scaling

- Each camera feeds an independent inference worker (Docker container).
- Kubernetes orchestrates worker scaling — add 100 cameras with no model changes.
- TimescaleDB partitions vehicle behaviour data by time to handle millions of records.

9.3 Integration with existing infrastructure

- REST API endpoints integrate with Bengaluru's existing e-challan portal.
- Evidence packets export directly as PDF challans.
- JHI data feeds the Smart City Operations Centre (SCOC) dashboard.
- The officer mobile app works offline and syncs when connectivity returns.

9.4 Privacy and legal compliance

- Plate data encrypted at rest (AES-256).
- Access to raw plate numbers restricted to authorised enforcement officers.
- Evidence-packet hash chain provides a tamper-proof audit trail for court use.
- Anonymised aggregate statistics available for urban planning and research.

SECTION 10 — EXPECTED OUTCOME DELIVERED

"A scalable AI-based traffic image analysis system capable of automatically identifying, classifying, and documenting traffic violations from photographic evidence, thereby reducing manual effort and improving the effectiveness of traffic monitoring and enforcement."

PRAHARI delivers this outcome across six measurable dimensions:

Outcome dimension	PRAHARI delivery	Measurable indicator
Scalable AI system	Docker-containerised, camera-agnostic, Kubernetes-ready	Handles 47+ junctions from one deployment
Automatic identification	YOLOv8 real-time detection, no human trigger	≥ 15 FPS on T4 GPU
Classification	7 violation classes + 3 offender tiers	mAP@0.5 ≥ 0.78 across classes
Documentation	Verifiable evidence packets with NL explanation	Every violation produces a signed record
Reducing manual effort	Auto patrol planning, auto JHI reporting, auto challan	Removes hours of daily analyst review per zone
Enforcement effectiveness	DNA scoring + optimizer + 48-hour forecast	More violations addressed per officer-hour (modelled)

In one sentence: PRAHARI treats the expected outcome's final clause — "improving the effectiveness of traffic monitoring and enforcement" — not as a side effect of detection, but as its primary design goal.

